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TYPE: Article CC:CCG

JOURNAL TITLE: Journal of intelligent information systems

USER JOURNAL TITLE: Journal of Intelligent Information Systems

ARTICLE TITLE: Leveraging personalized diversity level for recommendations with knowledge graph

ARTICLE AUTHOR: Ren, B., Yang, T., Ma, B. et al.

VOLUME: -

ISSUE: -

MONTH: -

YEAR: 2025

PAGES: -

ISSN: 1573-7675

OCLC #:

Processed by RapidX: 4/16/2026 1:26:01 AM

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Journal of Intelligent Information Systems
https://doi.org/10.1007/s10844-025-01013-8

RESEARCH

Leveraging personalized diversity level for recommendations with knowledge graph

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Received: 23 July 2025 / Revised: 19 November 2025 / Accepted: 21 November 2025
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Abstract
Recommender systems often face challenges related to information bias and the "filter bubble" effect, which limit content diversity and hinder user satisfaction. While much research has explored the balance between diversity and accuracy, traditional models often overlook the fact that different users have different levels of tolerance for diversity. In this work, we propose a novel approach PDRec-KG, which leverages knowledge graphs to enrich item representations with contextual information and integrate this with users' historical interactions to estimate their personalized diversity tolerance. Building upon this estimation, we develop a framework that dynamically adjusts the trade-off between diversity and accuracy to align with each user's preferences. Our method effectively tailors recommendations to individual diversity needs, offering a promising direction for enhancing both the quality and fairness of recommender systems. Comprehensive experiments on real-world datasets demonstrate the effectiveness of our approach. For instance, on the Last.FM dataset, PDRec-KG improves entity coverage by over 24% compared to the strongest baseline, while maintaining competitive accuracy, showcasing its ability to balance the trade-off between diversity and accuracy through personalized diverse recommendations.

Keywords Recommender system · Knowledge graph · Diversity

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Published online: 02 December 2025

1 Introduction

Recommender systems play a pivotal role in the contemporary Internet ecosystem (Mayer-Schönberger and Cukier, 2013), enabling users to discover and access content that aligns with their preferences. These systems are now integral to a diverse range of services(Wang et al., 2023a, b, 2022; Ying et al., 2018; Liu et al., 2022; Li et al., 2022), including social media, news platforms, music and video streaming, and e-commerce. Over the years, both industry and academia have achieved remarkable advancements in accuracy. For example, many recent approaches leverage knowledge graphs to better model complex relationships, thereby enhancing recommendation performance (Ma et al., 2025; Zhao et al., 2024; Zhang et al., 2024). Furthermore, current trends are exploring areas like federated learning for privacy-preserving recommendations (Di et al., 2025a, c, b, d), significantly enhancing user satisfaction and engagement.

However, despite these achievements, traditional recommender systems often rely heavily on historical user interactions and tend to prioritize content that is similar to users' past preferences. This emphasis on similarity, a core principle of many collaborative filtering and content-based methods, inherently creates problematic situations such as the ones characterized by the "filter bubble" effect (Ge et al., 2020), where users are repeatedly exposed to a narrow range of content, reinforcing their existing biases. Such an effect not only risks isolating users within informational silos but also limits their ability to encounter novel or serendipitous items outside their established interests, ultimately presenting them with an overly constrained and monotonous view of the world. This lack of diversity can lead to decreased long-term user engagement and limits the discovery of long-tail items. To address these adverse effects, researchers have increasingly shifted their focus beyond accuracy metrics to include diversity as a critical aspect of recommendation quality. For instance, approaches such as Maximal Marginal Relevance (MMR) (Di Noia et al., 2014) and greedy re-ranking (Ashkan et al., 2015) have been employed to enhance diversity in recommendation results. Moreover, recent advancements in Graph Neural Networks (GNNs) have focused on embedding generation techniques that better integrate diversity into the learned representations (Yang et al., 2023; Zheng et al., 2021).

Despite these efforts, many existing methods implicitly assume that all users exhibit a uniform tolerance for diversity. These approaches often aim to improve the overall diversity of recommendations at a system-wide level, overlooking the fact that user preferences for diversity can vary significantly (Wu et al., 2018; Eskandanian et al., 2017). For example, a recommendation list featuring classic science-fiction and action films might be well-suited for a user who primarily watches recent science-fiction movies, helping them explore a broader range of content. Conversely, the same list may appear insufficiently diverse to a user with eclectic tastes who has already watched films across various genres. This highlights the importance of defining and measuring individual users' tolerance for diversity, ensuring that recommendations not only align with their preferences but also respect their unique diversity acceptance levels.

To address this gap, many studies have started to focus on personalized diversity. For example, some studies explored post-processing techniques to reorder recommendation results based on users' diversity needs (Di Noia et al., 2014; Vargas et al., 2014). Another employed clustering-based methods, grouping users into clusters and tailoring the diversity of recommendations to the distinct requirements of each cluster (Eskandanian et al., 2017).

Although prior work has made progress in personalized diversity, most of these approaches operate at a coarse-grained level—either by improving only part of the recommendation pipeline or by focusing on group-level personalization. Moreover, they typically rely solely on category-level item information. Approaches that leverage richer item attributes to dynamically adjust the level of diversity throughout the entire recommendation process remain underexplored.

Our goal is to utilize fine-grained item information to comprehensively capture users' diversity preferences and incorporate their individual tolerance for diversity throughout the full recommendation pipeline. Unlike prior methods that depend solely on category-level data, we incorporate the rich contextual information provided by knowledge graphs. For instance, in the movie domain, knowledge graphs offer additional attributes such as directors, actors, production studios, and award histories. Leveraging these detailed item attributes allows us to extract nuanced user preferences for diversity from historical interactions with greater precision. Based on these insights, we introduce a novel framework, PDRec-KG. Its implementation involves three key stages: first, we model each user's personalized diversity tolerance by applying Shannon Entropy to their interaction patterns over KG entities and relations. Second, we propagate information through the knowledge graph using a diversity-oriented mechanism that learns user embeddings sensitive to less-frequent items. Finally, we employ a personalized re-ranking strategy that adjusts recommendation scores based on each user's pre-calculated diversity tolerance. This ensures truly fine-grained, individual-level personalized diversity in recommendation results.

In this work, we introduce a novel framework, denoted as PDRec-KG, which adapts recommendation diversity based on users' varying tolerance for diversity. By leveraging the rich contextual information in knowledge graphs, coupled with users' historical interaction data, we define and measure a diversity tolerance score for each user. This score informs a personalized recommendation strategy that dynamically adjusts diversity levels to align with individual preferences. Additionally, we introduce a diversity penalty mechanism, which considers the similarity between recommended items and users' past interactions. The penalty is modulated according to each user's diversity tolerance, ensuring a dual-layered balance between personalization and diversity. Advanced embedding techniques and conditional alignment methods further enhance the system's ability to integrate users' interests with the contextual richness of knowledge graphs, ensuring minimal compromise in accuracy.

The main contributions of this paper are summarized as follows:

- We propose a novel end-to-end framework, PDRec-KG, which for the first time quantifies and integrates personalized user diversity tolerance into the entire recommendation process. This is achieved through a novel method using Shannon Entropy on knowledge graph interactions, a diversity-aware training loss, and a personalized re-ranking mechanism.
- We design a diversity-oriented user embedding learning strategy that effectively mitigates the long-tail effect in user interactions, thereby enhancing the model's ability to recommend a wider variety of relevant items.
- We conduct extensive experiments on three real-world datasets, demonstrating that our model achieves statistically significant improvements in recommendation diversity while maintaining competitive accuracy compared to state-of-the-art baselines.

2 Related works

2.1 Recommender systems with knowledge graph

Knowledge graphs provide rich structured representations of entities (e.g., items, attributes) and their interrelations. By leveraging the comprehensive item information embedded in knowledge graphs, we can extract users' interest preferences from their historical interactions in a more detailed and holistic manner.

Many studies have explored incorporating knowledge graphs into recommendation systems to effectively utilize these detailed insights. For example, Deep Knowledge-Aware Network (DKN) (Wang et al., 2018) is a model for news recommendation that incorporates knowledge graph embeddings. It employs a knowledge-aware convolutional neural network (KCNN) (Kim, 2014) to integrate semantic and knowledge-level representations of news, using an attention mechanism to dynamically aggregate user interests based on their click history. MetaKRec (Wang et al., 2022) enhances recommendations by constructing collaborative Meta-Knowledge Graphs (Meta-KGs) to capture rich and heterogeneous semantic relationships among items.

KGAT (Wang et al., 2019) explicitly models high-order relations in collaborative knowledge graphs by combining user-item interactions with knowledge graph information. KGIN (Wang et al., 2021) introduces a graph neural network framework that explicitly models user intents and relational paths. By associating each user intent with a distribution over knowledge graph (KG) relations, KGIN captures the multi-intent nature of user-item interactions and ensures the independence of different intents. KGCL (Yang et al., 2022) addresses noise and sparsity issues in KGs by employing relation-aware knowledge aggregation, contrastive learning for denoising KG connections, and knowledge-guided augmentation for user-item interactions.

Most models that incorporate knowledge graphs into recommender systems primarily aim to utilize the rich item attribute information embedded in the graphs to better capture users' interest preferences, thereby improving the accuracy of recommendation results. However, these approaches largely overlook the trade-off between accuracy and diversity. Only a few models—such as KG-diverse (Liu et al., 2024)—have attempted to enhance recommendation diversity through knowledge graphs. Nevertheless, these methods typically adopt a uniform strategy from the perspective of user embedding to improve overall diversity, without accounting for the unique diversity preferences that can be extracted from knowledge graphs for different users. As a result, research on personalized diversity in such systems remains limited.

In this work, we leverage the rich contextual information in knowledge graphs to extract each user's distinct diversity preferences. We embed the notion of personalized diversity tolerance throughout the entire recommendation process and design individual-level diversity optimization strategies tailored to each user. This approach enables us to achieve personalized recommendations while effectively balancing accuracy and diversity.

2.2 Recommender systems for diversity

The goal of diversity-aware recommender systems is to broaden users' exposure by suggesting valuable items beyond their existing interests. Ziegler et al. (2005) first introduced

the concept of diversified recommendation and employed a greedy algorithm to generate a diversified Top-K recommendation list.

DPP (Chen et al., 2018) introduces a scalable approach for Maximum A Posteriori (MAP) inference to optimize the trade-off between relevance and diversity using a DPP kernel matrix constructed from item relevance scores and similarities. DGRec (Yang et al., 2023) improves recommendation diversity by identifying a diverse subset of neighbors for information aggregation through submodular function optimization. DivKG (Gan et al., 2020) learns KG embeddings using TransH (Wang et al., 2014) and designs a DPP kernel matrix to balance accuracy and diversity. DGCN (Zheng et al., 2021) integrates diversification into the candidate generation stage of recommendation systems, addressing the accuracy-diversity trade-off through rebalanced neighbor discovery, category-boosted negative sampling, and adversarial learning.

More recently, the emergence of Large Language Models (LLMs) has introduced a new paradigm for improving recommendation diversity. Leveraging their extensive world knowledge and reasoning capabilities, LLMs can be employed in re-ranking stages to enhance list-wise diversity (Carraro and Bridge, 2024). Other studies have evaluated their overall effectiveness, showing their potential but also highlighting challenges like computational cost and prompt sensitivity (Jiang et al., 2025).

Most existing efforts aimed at improving recommendation diversity primarily focus on enhancing overall diversity. Although some studies have recognized the importance of personalized diversity, they typically rely on re-ranking strategies (Vargas et al., 2014; Di Noia et al., 2014) or group-level personalization (Eskandanian et al., 2017), without deeply exploring the individualized needs of users. Consequently, diversity enhancement strategies in these models remain at a coarse-grained level. In addition, the majority of existing methods still rely solely on user-item interactions and category-level item information, further constraining the understanding of users' diversity preferences to a coarse level and limiting the ability to model diversity at the individual level comprehensively.

In contrast, our approach fully leverages the rich item information provided by knowledge graphs to model user interests and diversity tolerance with greater granularity. Moreover, we incorporate personalized diversity tolerance into all stages of the recommendation process. By integrating both diversity preference modeling and diversity enhancement strategies, our method enables more effective and in-depth individual-level diversity-aware recommendation.

3 Methodology

The overall architecture of the proposed **PDRec-KG** framework is shown in Fig. 1. It comprises three main modules: *(1) Diversity Tolerance Modeling*, *(2) Diversity-Oriented Knowledge Graph Propagation*, and *(3) Personalized Diversity Re-ranking*.

3.1 Problem formulation

User historical interaction graph Let $\mathcal{U}$ be the set of users ($|\mathcal{U}| = m$), and $\mathcal{I}$ be the set of items ($|\mathcal{I}| = n$). A bipartite interaction graph $\mathcal{G}_{HI} = \{\mathcal{V}, \mathcal{E}\}$ is constructed, where

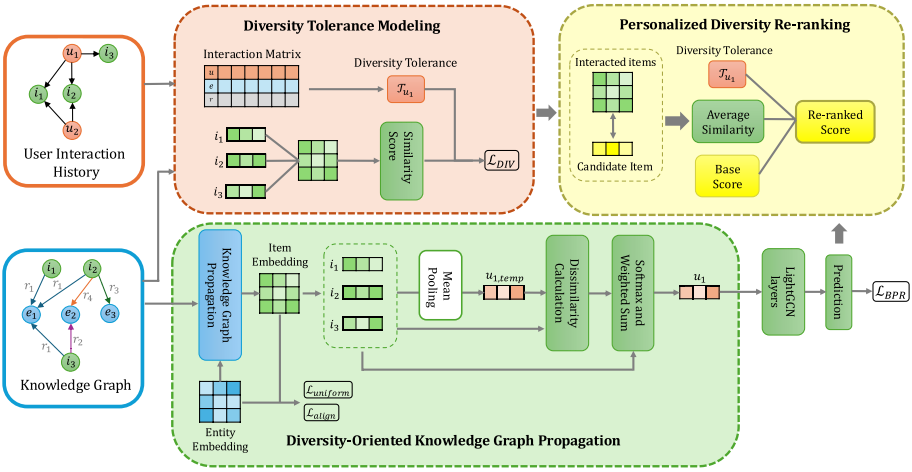


Fig. 1 Overall architecture of the proposed PDRec-KG model. The framework consists of three main stages: (1) Diversity Tolerance Modeling, which calculates a personalized tolerance score τ_u for each user from their historical interactions with KG entities and relations; (2) Diversity-Oriented Knowledge Graph Propagation, which learns user and item embeddings while mitigating the long-tail effect; and (3) Personalized Diversity Re-ranking, which adjusts final recommendation scores based on τ_u to tailor diversity for each user

$\mathcal{V} = \{\mathcal{U}, \mathcal{I}\}$ represents nodes and $\mathcal{E}$ represents edges. An edge (u, i) exists if user u has interacted with item i.

Knowledge Graph (KG) We further define a knowledge graph $\mathcal{G}_{KG} = \{(h, r, t)\}$, where each triplet consists of a head entity h, a relation r, and a tail entity t. The entity set $\mathcal{E}$ in the KG includes both items $\mathcal{I}$ and non-item entities. Thus, the KG encodes rich semantic information that potentially reveals deeper associations between items.

Task description Given $\mathcal{G}_{HI}$ and $\mathcal{G}_{KG}$, the goal is to recommend top-k items for each user, leveraging both collaborative signals and knowledge-graph semantics. Additionally, the model seeks to control and enhance recommendation *diversity* by incorporating user-specific preferences via **diversity tolerance** and a **diversity penalty**–based training strategy.

3.2 Diversity tolerance modeling

By integrating users' historical interaction records with the entities and relations associated with the items they have interacted with in the knowledge graph (KG), we can construct user-level interaction profiles at the entity and relation levels. Analyzing these interaction patterns reveals that users exhibit significantly different tendencies in how they engage with entities and relations. These behavioral tendencies reflect each user's tolerance for diversity in terms of the types of entities and relations they interact with.

For example, a user who listens to artists from many different genres (e.g., rock, jazz, classical) on Last.FM will have a higher entropy and thus a higher diversity tolerance score (τ_u). In contrast, a user who almost exclusively listens to heavy metal artists will have a lower entropy and a lower τ_u, indicating focused and narrow interests. These patterns indi-

cate that the scope and frequency of user interactions at the entity and relation levels can serve as meaningful indicators of their diversity tolerance.

To quantify this, we introduce Shannon Entropy as a measure of diversity tolerance based on the distribution of user interactions across entities and relations. Shannon entropy is a widely used metric in information theory that quantifies the uncertainty or unpredictability of a probability distribution. We compute the uncertainty of user interactions at the entity and relation levels based on their interaction patterns, and use this uncertainty as a proxy for representing the user's diversity tolerance.

Formally, the diversity tolerance of a user u is calculated as:

$$\tau_u = \frac{-\sum_x p_u(x)\ \log_2\big(p_u(x) + \epsilon\big)}{\log_2\big(|\mathcal{E}| + |\mathcal{R}|\big)}, \tag{1}$$

where $p_u(x)$represents the probability of interaction with an entity or relation x, $\mathcal{E}$ and $\mathcal{R}$ denote the sets of all entities and relations, and ϵ is a small constant for numerical stability.

Figure 2 illustrates the entropy distribution of user profiles in Last.FM dataset. This distribution with a particularly high variance highlights the significant differences in users' tolerance for diversity, underscoring the need for customized strategies tailored to users with varying tolerance levels to better meet their individual needs.

Diversity-aware training with loss adjustment To incorporate user diversity tolerance into the training process, we introduce a novel diversity loss term. This term dynamically adjusts recommendations based on each user's tolerance level. During training, we calculate the similarity between an item i and all previously interacted items $\mathcal{I}_u^{\text{hist}}$ for a user u. The similarity $S(i, \mathcal{I}_u^{\text{hist}})$ is defined as:

$$S(i, \mathcal{I}_u^{\text{hist}}) = \frac{1}{|\mathcal{I}_u^{\text{hist}}|} \sum_{j \in \mathcal{I}_u^{\text{hist}}} sim\big(\mathrm{e}_i, \mathrm{e}_j\big), \tag{2}$$

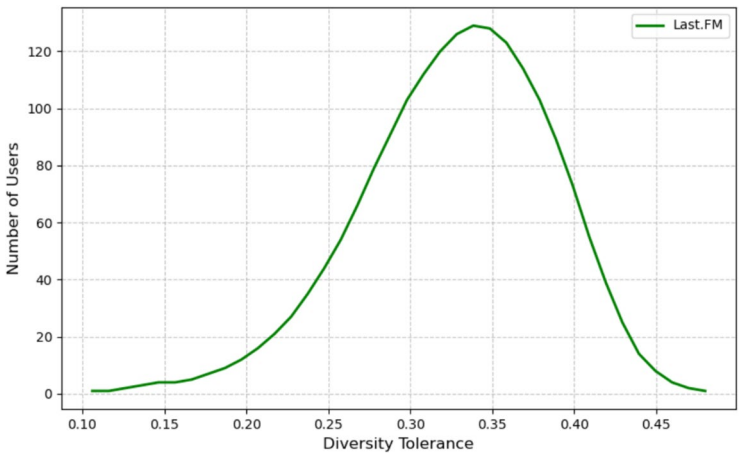


Fig. 2 Diversity tolerance distribution

where $sim(\mathrm{e}_i, \mathrm{e}_j)$ represents the cosine similarity between the embeddings e_i and e_j, formally defined as:

$$sim(\mathrm{e}_i, \mathrm{e}_j) = \frac{\mathrm{e}_i^\top \mathrm{e}_j}{\|\mathrm{e}_i\| \cdot \|\mathrm{e}_j\|}, \tag{3}$$

where $\|\mathrm{e}\|$ denotes the Euclidean norm of vector e.

The similarity score $S(i, \mathcal{I}_u^{\mathrm{hist}})$ is combined with the user's diversity tolerance τ_u to define the diversity-aware loss $\mathcal{L}_{\mathrm{div}}$:

$$\mathcal{L}_{\mathrm{div}} = -\frac{1}{|\mathcal{T}|} \sum_{(i,j)\in\mathcal{T}} \tau_u \cdot S(i, \mathcal{I}_u^{\mathrm{hist}}), \tag{4}$$

where $\mathcal{T}$ is the set of items in training process. This loss formulation allows the model to dynamically adjust recommendations based on each user's diversity tolerance. It is used to penalize recommendations that are overly similar to a user's history, with the strength of the penalty being proportional to the user's diversity tolerance τ_u. For users with a high tolerance, the penalty for similarity is stronger, encouraging the model to explore more diverse items. Conversely, for users with low tolerance, the penalty is weaker, allowing recommendations to remain closely aligned with their established preferences.

In addition to the diversity loss during training, we introduce a diversity penalty during the prediction phase, which further refines the system's ability to cater to users' varying diversity tolerances. This mechanism, described in detail in Section 3.5, ensures that the recommendations adapt to each user's diversity needs during inference.

Together, the diversity loss and penalty mechanisms enhance the overall diversity of recommendations while maintaining alignment with individual user preferences.

3.3 Diversity-oriented knowledge graph propagation

To address the need for capturing diverse user preferences while preserving recommendation accuracy, we propose an aggregation module that integrates entity-enriched item embeddings with a diversity-oriented user embedding learning mechanism.

3.3.1 Knowledge graph aggregation

The propagation begins by aggregating entity embeddings from the knowledge graph (KG). Let $\mathcal{E} \in \mathbb{R}^{n_e \times d}$ represent the entity embedding matrix, where n_e denotes the number of entities and d is the embedding dimension. Given the KG structure $\mathcal{G}_{KG} = \{(h, r, t)\}$, where h, r, and t represent the head entity, relation, and tail entity, respectively. We then aggregate the embedding of an item by treating the item as the head entity h and its associated entities as tail entity t at each KG layer. Formally:

$$\mathrm{e}_i^{(k)} = \frac{1}{|\mathcal{N}_h|} \sum_{t\in\mathcal{N}_h} \mathrm{e}_t^{(k-1)}, \tag{5}$$

where $\mathcal{N}_h$ denotes the set of tail entities connected to the item *i* via any relation *r*, e_t is the embedding of the tail entity *t* after $k-1$ layers aggregation, and $|\mathcal{N}_h|$ is the number of tail entities associated with *i*. This formulation captures the semantic relationships between entities and integrates contextual information from the KG.

To better utilize the rich entity information in the knowledge graph, we adopt a conditional alignment approach (Liu et al., 2024) to refine item embeddings from entity embeddings. Specifically, if two items i_1 and i_2 are connected to the same entities e_1 and e_2, their embeddings should align closely in the directions corresponding to e_1 and e_2. The conditional embeddings are denoted as:

$$e_{i|v_{\text{overlap}}} = e_i \odot e_{v_{\text{overlap}}}, \quad e_{j|v_{\text{overlap}}} = e_j \odot e_{v_{\text{overlap}}}, \tag{6}$$

where $e_{v_{\text{overlap}}} = \frac{1}{|\mathcal{O}_{i,j}|}\sum_{v\in\mathcal{O}_{i,j}} e_v$ is the average embedding of the overlapping entities between item i_1 and i_2, and $\mathcal{O}_{i,j} = \mathcal{N}_i \cap \mathcal{N}_j$ denote the set of overlapping entities between them. *alignment loss* is then given by

$$\mathcal{L}_{\text{align}} = \mathbb{E}_{(i,j)}\left[\left\|e_{i|v_{\text{overlap}}} - e_{j|v_{\text{overlap}}}\right\|_2^2\right]. \tag{7}$$

This design pulls *i* and *j* closer in embedding according to the shared context in Knowledge Graph.

Following (Yu et al., 2022), we further adopt a uniformity regularizer to ensure embeddings are uniformly scattered in the latent space:

$$\mathcal{L}_{\text{uniform}} = \log \mathbb{E}_{(i,j)} \exp\left(-2\left\|e_i - e_j\right\|_2^2\right). \tag{8}$$

Combining these two terms encourages items with overlapping entities to remain close while preserving an overall uniform distribution among item embeddings.

3.3.2 Diversity-oriented user embedding learning

Next, we compute an initial user embedding as a weighted average of item embeddings interacted by the user. Let $\mathcal{I}_u$ denote the set of items historically interacted with by user *u*, and let $\mathrm{e}_i^{\text{agg}}$ represent the aggregated embedding of item *i* obtained from the KG. The preliminary user embedding $\mathrm{u}_u^{\text{mean}}$ for user *u* is defined as:

$$\mathrm{e}_{u,\text{temp}}^{(k)} = \frac{1}{|\mathcal{I}_u|}\sum_{i\in\mathcal{I}_u} \mathrm{e}_i^{(k)}. \tag{9}$$

Traditional recommender systems refine user embeddings by weighting them based on the similarity to item embeddings, aligning user representations with frequently interacted items (Lu et al., 2023). While effective in amplifying the impact of such items, this approach often neglects the long-tail effect of item interactions.

Figure 3 demonstrates the distribution of item interaction frequencies in the MovieLens dataset. Items are ranked by interaction frequency (x-axis), and the y-axis shows the number of interactions per item. The figure reveals a pronounced long-tail effect: a small fraction of

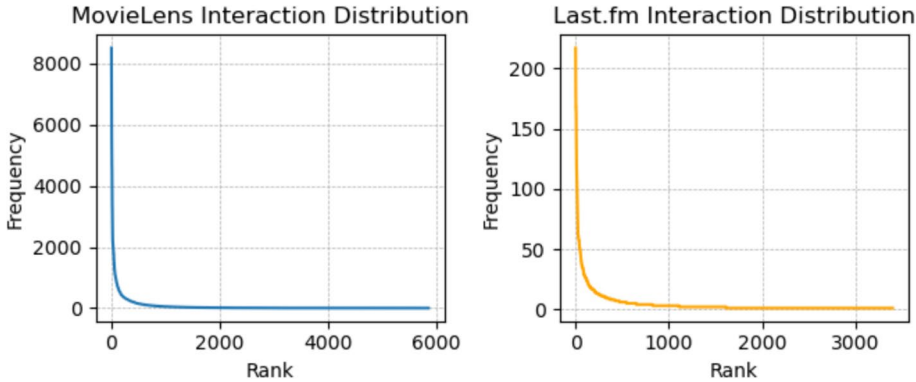


Fig. 3 Long-tail Effect in MovieLens & Last.FM dataset

items accounts for most interactions, while the majority are rarely interacted with. This bias skews user embeddings toward high-frequency items, limiting exposure to less frequent or novel items and reducing recommendation diversity.

To alleviate this, we employ a diversity-oriented embedding learning strategy that emphasizes items less similar to $e_u^{(k),\text{temp}}$hereby enhancing diversity. For a user u and interacted item i, the dissimilarity score $\text{diss}_{u,i}$ is calculated as follows:

$$\text{diss}_{u,i} = -\mathrm{e}_{u,\text{temp}}^{(k)\top} \mathrm{e}_i^{(k)}, \tag{10}$$

We then normalize the dissimilarity scores across all items interacted by u using a softmax function:

$$\alpha_{u,i} = \frac{\exp\left(diss_{u,i}\right)}{\sum_{j\in\mathcal{I}_u} \exp\left(diss_{u,j}\right)}. \tag{11}$$

Finally, the diversity-enhanced user embedding $\mathrm{e}_{u\,\text{div}}$ after k layers aggregation is computed as:

$$\mathrm{e}_{u\,\text{div}}^{(k)} = \sum_{i\in\mathcal{I}_u} \alpha_{u,i} \cdot \mathrm{e}_i^{(k)}. \tag{12}$$

This approach mitigates the long-tail effect by reducing the influence of excessively similar items, thereby enhancing the diversity of user embeddings. Furthermore, previous research (Wang et al., 2019, 2021) shows that stacking multiple layers effectively captures higher-order knowledge graph information. Thus, we stack user and item embeddings across all k layers to construct the final embeddings:

$$\mathrm{e}_u = \mathrm{e}_{u,\text{div}}^{(0)} + \ldots + \mathrm{e}_{u,\text{div}}^{(k)}, \quad \mathrm{e}_i = \mathrm{e}_i^{(0)} + \ldots + \mathrm{e}_i^{(k)}. \tag{13}$$

3.4 Prediction and optimization

After k rounds of KG propagation and user embedding construction, we incorporate *Light Graph Convolution* (He et al., 2020) (LightGCN) layers to capture collaborative signals from user historical interactions. The Light Graph Convolution (LightGCN) framework leverages user-item interaction information to propagate embeddings through a simplified graph convolution mechanism. The propagation rules for user and item embeddings across the l-th layer are defined as follows:

$$e_u^{(l)} = \sum_{i \in \mathcal{H}_u} \frac{1}{\sqrt{|\mathcal{H}_u|}\sqrt{|\mathcal{N}_i|}} e_i^{(l-1)}, \tag{14}$$

$$e_i^{(l)} = \sum_{u \in \mathcal{N}_i} \frac{1}{\sqrt{|\mathcal{N}_i|}\sqrt{|\mathcal{H}_u|}} e_u^{(l-1)}, \tag{15}$$

where the sets $\mathcal{H}_u$ and $\mathcal{N}_i$ denote the items historically interacted by user u and the users who have interacted with item i, respectively. Meanwhile, $e_u^{(0)}$ and $e_i^{(0)}$ are the embedding obtained after knowledge graph propagation. Then, similar to Knowledge Graph Propagation, we integrate the embeddings from each layer into the final embeddings:

$$e_u = \frac{1}{L}\sum_{l=0}^{L} e_u^{(l)}, \quad e_i = \frac{1}{L}\sum_{l=0}^{L} e_i^{(l)}. \tag{16}$$

A fter we gained the final user and item embedding, to learn the underlying embeddings, we adopt the Bayesian Personalized Ranking (BPR) objective (Rendle et al., 2012) for recommendation:

$$\mathcal{L}_{\mathrm{BPR}} = -\sum_{(u,i,j)\in\mathcal{O}} \log \sigma\Big(\big(e_u^\top e_i\big) - \big(e_u^\top e_j\big)\Big), \tag{17}$$

Here, i is an observed item in $\mathcal{H}_u$, and j is an unobserved item.

Overall objective Finally, the overall loss function is:

$$\mathcal{L} = \mathcal{L}_{\mathrm{BPR}} + \mathcal{L}_{\mathrm{div}} + \mathcal{L}_{\mathrm{align}} + \mathcal{L}_{\mathrm{uniform}} + \lambda \left\|\Theta\right\|_2^2. \tag{18}$$

3.5 Personalized diversity re-ranking

To further refine the final recommendation list according to each user's diversity tolerance, we apply a post-processing *re-ranking* step. Let $s_{u,i}$ be the base recommendation score for user u and item i obtained from PDRec-KG. We denote the user's diversity tolerance by $\tau_u \in [0, 1]$.

For each item i, we compute a *diversity penalty* by measuring its similarity to the user's historically interacted items. Formally, let $\mathcal{H}_u$ be the set of items that user u has interacted with, and we adopt cosine similarity function to measure the similarity between the item embeddings e_i and e_j. We extract the top K most similar historical items of u (denoted $\mathcal{K}_u \subseteq \mathcal{H}_u$) and define the diversity penalty $\mathcal{P}(i, \mathcal{H}_u)$ as:

$$\mathcal{P}(i, \mathcal{H}_u) = \frac{1}{|\mathcal{K}_u|} \sum_{j \in \mathcal{K}_u} \frac{\mathrm{e}_i^\top \mathrm{e}_j}{\|\mathrm{e}_i\| \|\mathrm{e}_j\|}, \tag{19}$$

where $\mathcal{K}_u$ includes the top K historically interacted items of u that are most similar to i.

We then adjust the original score $s_{u,i}$ by subtracting a penalty term scaled by user diversity tolerance and a coefficient μ. Concretely, the *re-ranked* score $\hat{s}_{u,i}$ is given by:

$$\hat{s}_{u,i} = s_{u,i} - \tau_u \cdot \mathcal{P}(i, \mathcal{H}_u) \cdot \mu, \tag{20}$$

All candidate items for user u are then ranked by $\hat{s}_{u,i}$. By incorporating a user's τ_u into the final re-ranking, the system dynamically balances exploration and exploitation. Users with high diversity tolerance receive more diverse recommendations (stronger penalty for highly similar items), while users with low diversity tolerance receive relatively less diverse recommendations (weaker penalty). This step helps ensure that each user's individual preference for diversity is systematically accounted for in the final ranked list.

4 Experiments

In this section, we conduct empirical studies to evaluate the effectiveness of PDRec-KG. The results of the experiments answer the following research questions:

- **RQ1:** How does PDRec-KG perform compared to other recommendation methods?
- **RQ2:** What is the effect of each proposed component in PDRec-KG?
- **RQ3:** What are the impacts of hyper-parameters: number of KG propagation layers L and diversity penalty coefficient μ?
- **RQ4:** What is the impact of personalized diversity tolerance on recommendation outcomes?

4.1 Experiment setup

4.1.1 Datasets

We conducted experiments on three widely-used (Wang et al., 2023, 2019, 2021) real-world datasets: Last.FM (Bertin-Mahieux et al., 2011), MovieLens (Harper and Konstan, 2015), and Amazon-Book (He and McAuley, 2016). Following prior studies, we split each user's historical interactions into training, validation, and test sets with a ratio of 8:1:1. Detailed statistics of these datasets are presented in Table 1.

Table 1 Dataset Statistics

Dataset	Amazon-Book	Last.FM	Movielens
#Users	70,679	1,872	37,385
#Items	24,915	3,846	6,182
#Interactions	846,434	42,346	539,300
#Entities	113,487	9,366	24,536
#Relations	39	60	20
#Triplets	2,557,746	15,518	237,155

4.1.2 Baseline models

To comprehensively assess the performance of our model, we selected several representative baseline models, categorized into four groups:

1. General Recommender Systems:

   - MF (Koren et al., 2009): A collaborative filtering method that factorizes the user-item interaction matrix into latent factors to capture linear relationships.
   - LightGCN (He et al., 2020): A simplified graph-based model using only neighborhood aggregation to efficiently learn high-order user-item connections.

2. Diversity-aware Graph-based Recommender Systems:

   - DGCN (Zheng et al., 2021): A graph convolutional network that incorporates diversity constraints to balance accuracy and diversity.
   - DGRec (Yang et al., 2023): A graph neural network that enhances diversity by selecting diverse neighbors using a submodular function.

3. Knowledge Graph-enhanced Recommender Systems:

   - KGAT (Wang et al., 2019): An attention-based model leveraging entity influence in knowledge graphs for better recommendations.
   - KGIN (Wang et al., 2021): A model using relational path-aware aggregation to refine user and item embeddings via relation information.
   - KGIE (Li et al., 2024): A model that interactively enhances entity representations within the knowledge graph.
   - KGCCL (Meng et al., 2024): A knowledge graph contrastive collaborative learning framework.

4. LLM-based Re-ranking Systems:

   - MF+Gemma3 & MF+Qwen3: To evaluate the capabilities of LLMs in a manner compatible with our data structure, we employ them as re-rankers. First, the base MF model generates a list of candidate items for each user. Then, an LLM (Gemma-3 or Qwen-3) is prompted with the user's interaction history and the candidate list, and tasked to re-rank these items. This two-stage approach allows us to leverage the semantic understanding of LLMs for list personalization. Due to the

high computational cost and the need for item title information, these models are evaluated only on the Movielens dataset.

4.1.3 Evaluation metrics

In our experiments, we evaluate accuracy and diversity, respectively.

Accuracy Metrics. To assess accuracy, we employ the widely adopted metrics Recall@K and NDCG@K (Krichene and Rendle, 2020). These metrics evaluate the effectiveness of the top-K recommendations, where K is set to 20 and 40 in our experiments. Recall@K measures the proportion of relevant items successfully recommended within the top-K results, while NDCG@K incorporates the ranking quality, assigning higher weights to relevant items appearing at higher ranks.

Diversity Metrics. To evaluate diversity, a commonly used metric is Intra-List Distance (ILD) (Smyth and McClave, 2001; Ziegler et al., 2005), which measures the dissimilarity among items within a recommendation list. While ILD offers valuable insights into item-level diversity based on pairwise similarity, it primarily operates on surface-level features (e.g., genres or content embeddings) and fails to fully capture the semantic richness embedded in structured knowledge graphs.

To address this limitation, we adopt two graph-aware diversity metrics—Entity Coverage (EC) and Relation Coverage (RC)—inspired by recent work (Liu et al., 2024). These metrics are specifically designed to exploit the multi-dimensional semantics of knowledge graphs, providing a more fine-grained evaluation of diversity across structural and relational dimensions:

Entity Coverage (EC): Measures the number of distinct entities associated with the items in the recommendation list.

Relation Coverage (RC): Measures the number of unique relations involved in the recommended items.

Higher EC and RC values indicate broader diversity, as they reflect the extent to which the recommendations span different entities and relations in the knowledge graph. We compute these metrics as the proportion of unique entities and relations covered in a recommendation list relative to the total number present in the knowledge graph.

4.1.4 LLM Re-ranking prompt

For the LLM-based re-ranking baselines (MF+Gemma3 and MF+Qwen3), we utilized a unified prompt template to guide the models. The prompt was designed to provide clear context and instructions for re-ranking the candidate item list generated by the MF model, structured as follows:

```
<Instructions>
Considering a user:{user_profile_summary}.

Here is the movies this user has watched:
{user_history_titles}.

Your task is to rerank the candidate list and provide a final
top-{k} recommendation list where the goal is to balance
relevance and diversity.

<Candidate List>
{candidate_movie_titles}

<Output Format>:
Strictly use the following format for the output, and don't
provide additional information.
1-> <item name>
2-> <item name>
...
{k}-> <item name>
```

Prompt 1: The LLM re-ranking prompt template. Tags in angular brackets are for exposition and do not appear in the final prompt. Placeholders for data are enclosed in braces.

In this template, the *user profile summary* placeholder was replaced with a structured textual summary of the user's most watched genres and preferred release years. The *user history titles* and *candidate movie titles* placeholders were replaced with the user's actual viewing history and the candidate movie list generated by the MF model, respectively.

4.2 Performance comparison (RQ1)

Parameter Setting. We implemented PDRec-KG and all baseline models using PyTorch, utilizing the Adam optimizer (Kingma and Ba, 2015) for training. All models were trained until no performance improvement was observed on the validation set for 10 consecutive epochs. Following the setting of KGAT (Wang et al., 2019), we fix the embedding size to 64 for all models. Grid search is performed to tune the hyper-parameters for each method. The model parameters are initialized by Xavier initializer. Table 2 presents a comprehensive performance comparison between PDRec-KG and the baseline models. The results highlight the following observations:

- PDRec-KG outperforms all baseline models on most diversity metrics across all datasets, with the exception of relation coverage on the MovieLens dataset. Interestingly, the relation coverage of almost all models on MovieLens is similar. This can be attributed to the dataset containing only 20 distinct relations, making it possible for even a small recommendation set to encompass most relations.
- Our model achieves the highest Entity Coverage across all datasets while maintaining minimal trade-offs in accuracy. For example, compared to the LightGCN model, PDRec-KG demonstrates comparable performance in terms of Recall and NDCG on the Amazon-Book and MovieLens datasets. This demonstrates the effectiveness of our

model in balancing accuracy and diversity.

- When compared to models like DGCN and DGRec, which are specifically designed to enhance diversity, PDRec-KG not only achieves superior performance on diversity metrics but also surpasses these baselines in accuracy metrics. This highlights the strength of PDRec-KG in leveraging the rich semantic and contextual information provided by the knowledge graph. By accurately modeling user interests and preferences, PDRec-KG delivers more diverse recommendations without compromising recommendation accuracy.
- Compared to other knowledge-based models(KGIN, KGAT, KGIE and KGCCL), which also leverage the rich semantic information in knowledge graphs to generate user and item embeddings, our model demonstrates slightly lower accuracy than KGIN and achieves comparable performance on KGAT, KGIE and KGCCL. As for diversity metrics, our model successfully surpasses these baselines. This indicates that while utilizing knowledge graphs similarly, our model effectively incorporates personalized diversity tolerance to balance the varying diversity preferences of different users. As a result, it achieves a substantial improvement in overall diversity without sacrificing too much relevance.
- The LLM-based re-ranking experiments also offer valuable insights. As shown in Table 2, both Gemma3 and Qwen3 fail to improve upon the base MF model on the Movielens dataset, with Qwen3 showing a particularly sharp decline in performance. This suggests that simply applying LLMs for diversity re-ranking is highly sensitive to the chosen model and prompting strategy, and may not inherently outperform well-established collaborative filtering or GNN-based methods without careful tuning and design.

To validate the significance of our model's diversity improvements, we performed a paired, two-tailed t-test. For each key diversity metric (e.g., EC@20), we compared the list of per-user scores generated by PDRec-KG against the list generated by the strongest baseline. The results confirm that the improvements in Entity Coverage and Relation Coverage where our model excels are statistically significant ($p < 0.05$), indicating that the observed gains are not due to random chance.

4.3 Ablation study (RQ2)

We conducted an ablation study on Last.FM dataset to evaluate the contribution of the key components of PDRec-KG. Specifically, we examined the impact of: (1) removing diversity tolerance modeling and personalized diversity re-ranking (DTPR) module, (2) removing the diversity-oriented user embedding learning module (DUEL), and (3) removing information propagation from the knowledge graph layers (KGP). Table 3 summarizes the accuracy and diversity performance of PDRec-KG and its ablated variations.

- Without diversity tolerance modeling and personalized diversity re-ranking (DTPR) module, the system no longer calculates users' tolerance for diversity, nor does it individually adjust the recommendation process based on each user's diversity tolerance. As a result, all diversity metrics experienced a significant decline. Entity Coverage decreased by 7.40% (from 152.63 to 141.32), and Relation Coverage dropped by 5.23% (from 16.95 to 16.06). In contrast, accuracy metrics showed a slight improvement of

Table 2 Performance comparison on the Amazon-Book, Last.FM, and Movielens-1M datasets

Dataset	Method	Accuracy				**Diversity**			
		Recall@20	Recall@40	NDCG@20	NDCG@40	EC@20	EC@40	RC@20	RC@40
Amazon-Book	MF	0.0978	0.1432	0.0519	0.0640	63.0985	96.9736	20.1242	21.6758
	LightGCN	0.1103	0.1684	0.0609	0.0717	64.1004	95.2032	20.0031	21.9493
	DGCN	0.0813	0.1307	0.0412	0.0517	56.0879	84.4123	19.6245	21.4578
	DGRec	0.0931	0.1424	0.0487	0.0596	65.2413	97.6023	20.2001	22.0023
	KGAT	0.1251	**0.1833**	0.0611	0.0756	52.3986	78.8934	18.2374	19.8943
	KGIN	**0.1297**	0.1812	**0.0682**	**0.0796**	50.1243	75.3122	18.1874	20.3402
	KGIE	0.1010	0.1513	0.0521	0.0648	64.3163	96.7607	20.4481	22.5416
	KGCCL	0.1021	0.1555	0.0535	0.0668	65.1563	97.8710	20.6263	21.7071
	PDRec-KG (Ours)	0.1019	0.1548	0.0513	0.0647	**67.0420**	**100.0150**	**20.7544**	**22.6288**
Last.FM	MF	0.3302	0.4356	0.1795	0.2123	68.2258	122.5086	11.5234	14.3156
	LightGCN	**0.3691**	**0.4709**	**0.2023**	**0.2267**	67.0934	117.1245	10.5367	13.3321
	DGCN	0.2338	0.3503	0.1256	0.1482	59.7983	114.2347	10.8685	13.7845
	DGRec	0.2617	0.3634	0.1302	0.1498	60.3327	112.0213	11.2236	14.1674
	KGAT	0.3211	0.4402	0.1834	0.2156	65.9832	119.0126	11.1432	14.1256
	KGIN	0.3423	0.4521	0.1622	0.1903	66.1124	118.0214	11.9201	14.4723
	KGIE	0.3259	0.4337	0.1743	0.1991	70.7267	122.2489	11.6478	14.1021
	KGCCL	0.3267	0.4291	0.1812	0.2050	67.9028	123.0729	11.2348	14.0148
	PDRec-KG (Ours)	0.3413	0.4532	0.1685	0.1945	**88.2673**	**152.6288**	**14.6228**	**16.9467**
Movielens-1M	MF	0.4065	0.5297	0.2187	0.2490	454.3015	805.9360	17.1941	17.3594
	LightGCN	0.4157	0.5462	0.2236	0.2557	452.7703	803.3682	17.1809	17.3426
	DGCN	0.3625	0.4941	0.1874	0.2198	444.5471	799.3097	**17.2358**	**17.4053**
	DGRec	0.3676	0.4981	0.1917	0.2238	445.0291	798.6214	17.2310	17.3980
	KGAT	0.4113	0.5278	0.2201	0.2511	445.1102	793.3092	17.2101	17.3814
	KGIN	**0.4381**	**0.5772**	**0.2503**	**0.2823**	448.1276	800.2175	17.2284	17.3321
	KGIE	0.3737	0.4911	0.2012	0.2300	449.7388	799.9068	17.1735	17.3541
	KGCCL	0.3897	0.5129	0.2091	0.2393	453.8744	805.9916	17.1803	17.3409
	MF+Qwen3*	0.1600	0.3900	0.0506	0.0977	435.2126	788.5741	17.2212	17.4031

Table 2 (continued)

Dataset	Method	Accuracy				**Diversity**			
		Recall@20	Recall@40	NDCG@20	NDCG@40	EC@20	EC@40	RC@20	RC@40
	MF+Gemma3*	0.3950	0.5200	0.1755	0.2030	438.8715	796.7142	17.2132	17.4014
	PDRec-KG (Ours)	0.4027	0.5401	0.2122	0.2460	**463.4147**	**812.0994**	17.1110	17.2952

The best results are highlighted in **bold**, and the second-best results are underlined. *LLM-based models were evaluated only on the Movielens-1M dataset

Table 3 Ablation study results on the Last.FM dataset

Method	R@40	N@40	EC@40	RC@40
PDRec-KG	0.4532	0.1945	**152.6288**	**16.9467**
w/o DTPR	**0.4616**	0.2051	141.3173	16.0603
w/o DUEL	0.4479	0.1982	136.9467	15.7771
w/o KGP	0.4508	**0.2114**	121.7929	14.0586

1.85% (R@40 increased from 0.4532 to 0.4616). These results demonstrate that the personalized diversity module effectively captures users' varying diversity requirements, substantially improving overall system diversity without a notable sacrifice in accuracy.

- Removing the diversity-oriented user embedding learning module (DUEL) reduced user embeddings to a simple average of the embeddings of interacted items. This adjustment caused minimal changes in accuracy metrics (R@40 decreased slightly from 0.4532 to 0.4479) but led to significant declines in diversity metrics, with Entity Coverage decreasing by 10.29% (from 152.63 to 136.95). These findings indicate that the diversity embedding smoothing module plays a crucial role in incorporating diverse information into user embeddings, enhancing the model's ability to deliver varied recommendations.
- Without knowledge graph propagation (KGP), the system relied solely on the user-item interaction graph to learn item and user embeddings. This simplification resulted in substantial reductions in diversity metrics: Entity Coverage dropped by 20.20% (from 152.63 to 121.79), and Relation Coverage decreased by 17.07% (from 16.95 to 14.06). These results highlight the importance of knowledge graph propagation in enriching the contextual semantic information available to the model, thereby enhancing its capacity to capture users' nuanced interests and deliver more diverse content.

To further validate the impact of each component, we conducted a paired, two-tailed t-test on the ablation results. Specifically, we compared the per-user diversity scores of the full PDRec-KG model against each of its ablated variants. The removal of the DTPR, DUEL, and KGP modules each resulted in a statistically significant decrease in both Entity and Relation Coverage ($p < 0.05$), confirming that each component provides a meaningful contribution to the model's performance.

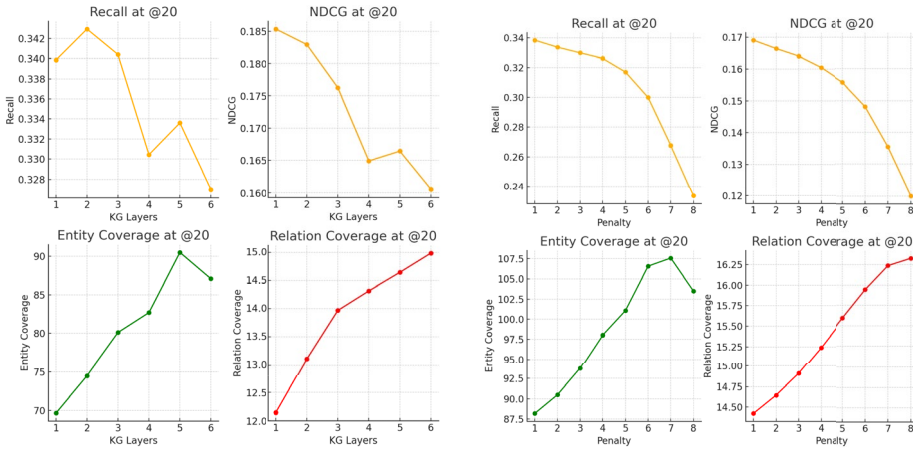
4.4 Parameter sensitivity (RQ3)

To gain deeper insights into the structure and effectiveness of PDRec-KG, we evaluated the influence of two key hyperparameters on the LastFM dataset: the number of KG propagation layers L and the diversity penalty coefficient.

4.4.1 KG Propagation Layers (L)

We varied the number of KG propagation layers from 1 to 6 to assess their impact on accuracy and diversity metrics. Figure 4(a) illustrates the results.

For diversity metrics, overall diversity improves as the number of KG propagation layers increases. This demonstrates that PDRec-KG effectively leverages information from the knowledge graph to enhance diversity. However, a notable decline in Entity Coverage is observed when the number of layers exceeds 5. This decrease can likely be attributed to the



(a) KG Propagation Layers (b) Penalty Coefficient

Fig. 4 Parameter sensitivity of KG propagation depth and diversity penalty

reduced discriminative power caused by excessive stacking of propagation layers, leading to over-smoothing of entity representations.

In terms of accuracy metrics, the overall accuracy decreases with an increasing number of layers. This decline can be explained by the over-smoothing problem, where deeper propagation layers cause node embeddings to become indistinguishable, thereby reducing the model's ability to capture nuanced user-item interactions.

4.4.2 Diversity penalty coefficient (μ)

We varied the diversity penalty coefficient from 1 to 8 to evaluate its impact on diversity and accuracy metrics. The results are illustrated in Fig. 4(b).

As the diversity penalty coefficient increases, the system's diversity metrics show an overall upward trend. However, when the coefficient exceeds 7, Entity Coverage begins to decline. This decline may indicate a threshold beyond which excessive diversity leads to disordered recommendations. For accuracy metrics, a slight decline is observed as the diversity penalty coefficient increases between 1 and 4. This mild reduction suggests that the system effectively balances diversity and accuracy, successfully recommending items aligned with users' personalized tolerance for diversity while achieving overall improvements in diversity. However, when the coefficient exceeds 5, the accuracy declines sharply. This indicates that over-prioritizing diversity compromises alignment with user preferences and recommendation quality.

4.5 Group-wise analysis (RQ4)

Inspired by previous study (Eskandanian et al., 2017), we adopted a clustering-based approach to segment users in the Last.FM dataset into four groups based on their diversity tolerance. This allowed us to analyze the significance and importance of personalized diversity recommendations tailored to each user's diversity tolerance level.

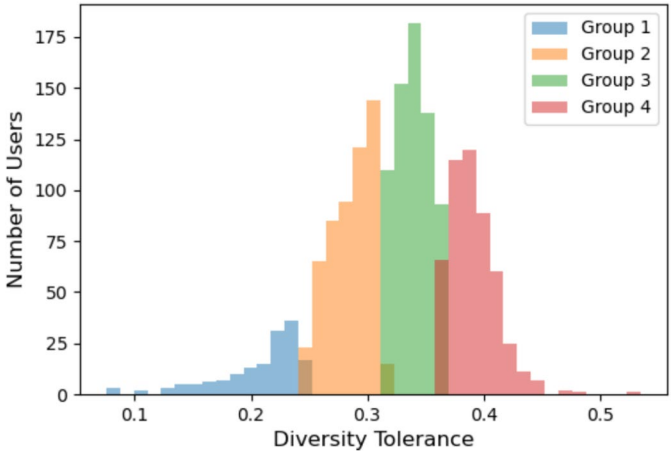


Fig. 5 User groups based on diveristy tolerance

Table 4 Group-wise analysis and recommendation results (@20)

Group	Recall	nDCG	RC	EC	Average		
					DT	Relations	Entities
Group 1	0.2944	0.1354	14.5547	78.0656	0.2067	7.56	7.18
Group 2	0.3183	0.1707	14.5486	78.0411	0.2857	13.24	13.22
Group 3	0.3219	0.1561	14.9420	91.8841	0.3396	18.88	24.73
Group 4	0.3555	0.1718	15.0887	105.9133	0.3911	26.26	50.88
Overall-DGCN	0.2338	0.1256	10.8685	59.7983	–		
Overall-DGRec	0.2617	0.1302	11.2236	60.3327	–		

The "Average" columns represent statistics of the user groups

Using k-means clustering, we grouped users into four clusters and analyzed the distribution of diversity tolerance scores and the number of users in each group, as shown in Figure 5. For each group, we report the average diversity tolerance (DT), the number of interacted relations and entities, and the recommendation performance of PDRec-KG using accuracy metrics (Recall, NDCG) and diversity metrics (Relation Coverage, RC; Entity Coverage, EC) in Table 4. For comparison, Table 4 also includes the overall results of two diversity-focused baselines, Overall-DGCN and Overall-DGRec.

Users in different clusters exhibit distinct interaction behaviors, aligned with their diversity tolerance. Those with broader interests interact with significantly more entities and relations than those with narrower interests. This supports our hypothesis that a single recommendation list may imply vastly different levels of diversity for different users. This variation underscores the need for diversity enhancements tailored to users' specific preferences, as overall diversity improvements may fail to meet individual needs.

Performance metrics show that recommendation diversity increases with users' diversity tolerance, demonstrating that our system effectively captures users' varying diversity preferences and provides personalized recommendations aligned with these preferences. Additionally, accuracy improves for users with higher diversity tolerance, likely because our system effectively balances accuracy and diversity by leveraging the knowledge graph.

For users with lower diversity tolerance, our system mitigates the filter bubble effect by introducing some diversity, which may slightly impact accuracy.

Notably, when compared to the overall results of the two baseline models aimed at improving diversity, our model demonstrates superior performance. Even for the group with lower diversity tolerance, accuracy metrics surpass the two baselines, while diversity metrics are significantly higher across all user groups. This demonstrates that leveraging rich attributes information from the knowledge graph and individual-level diversity enhancements can deliver content aligned with users' diversity tolerance levels without sacrificing accuracy, satisfying broad-interest users while mitigating filter bubble effects for focused-interest users.

5 Conclusion and future work

In this paper, we propose PDRec-KG, a model based on knowledge graphs and user interaction history to provide recommendations with personalized levels of diversity. Specifically, we define users' diversity tolerance by extracting user-item interaction information at the entity and relation levels from the knowledge graph and user interaction data. This information is used to measure users' demand for diversity and to personalize recommendation results based on their diversity tolerance. Additionally, we introduce diversity-oriented user embedding learning, which enables embeddings derived from users' historical interactions to mitigate the influence of the long-tail effect. This approach enhances the model's ability to capture information about niche and less popular items, thereby improving overall diversity performance. We evaluate our model on multiple datasets, demonstrating its capability to effectively identify users' diverse needs and provide recommendations tailored to varying levels of diversity without compromising accuracy.

For the limitation, currently our model measures users' tolerance for diversity based on their overall interactions with entities and relations, however, entities and relations can also be further categorized based on their internal relevance, enabling a more granular understanding of users' diversity preferences within these categories. In the future, we plan to explore this direction to achieve more fine-grained personalization.

Furthermore, we intend to investigate dynamic models that can capture how a user's diversity preference might evolve over time or differ across sessions. Finally, our initial experiments with LLM-based re-rankers highlight a promising but challenging direction; future work could focus on developing more sophisticated hybrid models that effectively combine our graph-based structural approach with the rich semantic reasoning of Large Language Models. To validate these future explorations, we also plan to conduct comprehensive user studies to more directly assess and deliver individual-level diversity recommendations.

Author Contributions Conceptualization: Baofeng Ren, Boxuan Ma; Methodology: Baofeng Ren, Tianyuan Yang; Software: Baofeng Ren, Tianyuan Yang; Validation: Boxuan Ma; Resources: Baofeng Ren; Writing – original draft preparation: Baofeng Ren; Writing – review and editing: Shin'ichi Konomi; Supervision: Shin'ichi Konomi.

Funding This work was supported by JSPS KAKENHI Grant Numbers JP20H00622 and JP24K20903. It was also supported by JST SPRING, Grant Number JPMJSP2136.

Data Availability The datasets used in this study (Amazon-Book, Last.FM, and Movielens) are publicly available and can be obtained from their respective original sources as cited in the text. The source code used in this study is openly available at https://github.com/renbf41/PDRec-KG.

Declarations

Competing Interests The authors declare no competing interests.

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